3.5 Multilinear Algebra

3.5.1 Multilinear Maps and Determinants

Remark 3.5.1
The purpose of this section is to define the determinant, rather than develop any further theory. The approach is a little non-standard as we take an abstract module-theoretic approach as in [Bou98a], but avoid use of the tensor algebra. The connection is there are natural isomorphisms (i.e. the tensor algebra represents the functor $L^n_a(M;-)$)

$$L^n_a(M;N) \xrightarrow{\sim} \mathrm{Hom}(\Lambda^n(M), N)$$

and

$$\Lambda^n(M)^\vee \xrightarrow{\sim} \Lambda^n(M^\vee)$$

when M is finite free. Therefore the exterior product we define really corresponds to the exterior product on (the tensor algebra of) $M^\vee$.

Definition 3.5.2 (Multilinear Map)
Let $M_1, ..., M_n, N$ be A-modules then a map

$$\psi : M_1 \times ... \times M_n \to N$$

*is A-**multilinear** if it is A-linear in each variable, whilst fixing the other variables at any value.*

Definition 3.5.3 (Bilinear form)
*Let M, N be A-modules then $\psi : M \times N \to A$ is a **bilinear form** if it is A-multilinear.*

Denote the set of such bilinear pairings by $\mathrm{Bilin}_A(M,N)$. It is naturally an A-module.

Proposition 3.5.4
Let M, N be A-modules then there is a natural bijection

$$\begin{aligned} \mathrm{Hom}_A(M, \mathrm{Hom}_A(N,A)) &\leftrightarrow \mathrm{Bilin}_A(M,N) \leftrightarrow \mathrm{Hom}_A(N, \mathrm{Hom}_A(M,A)) \\ \psi_L &\leftarrow \psi \\ \psi &\qquad\qquad \to \psi_R \end{aligned}$$

where

$$\psi_L(m)(n) = \psi(m,n) = \psi_R(n)(m)$$

Definition 3.5.5 (Alternating map)
*An A-multilinear map $f : M^n \to N$ is **alternating** if*

$$f(x_1, ..., x_n) = 0$$

whenever $x_i = x_{i+1}$ for some $i = 1..., n-1$.

*Denote by $L^n_a(M,N)$ the set of such alternating maps, and $L^n_a(M) := L^n_a(M,A)$ the set of **alternating forms**. These are clearly A-modules.*

Proposition 3.5.6 (Functorial Properties)
Let M be an A-module and $L^k_a(M)$ the set of k-alternating forms. Then it is a contravariant functor in M, that is if $g : M \to N$ then there is a well-defined map

$$\begin{aligned} L^k_a(g) : L^k_a(N) &\to L^k_a(M) \\ \psi &\to \psi \circ g^{(k)} \end{aligned}$$

such that $L^k_a(g \circ h) = L^k_a(h) \circ L^k_a(g)$

Lemma 3.5.7
Let M be an A-module and $\psi \in L^{p-1}_a(M;A)$. Then

$$\psi'(x_1, ..., x_p) := \sum_{j=1}^{p} (-1)^{j+1} \psi(x_1, ..., \hat{x_j}, ..., x_p) x_j$$

is an element of $L^p(M;M)$.

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Proof. It is clear that ψ' is multilinear, so we only need to show it is alternating. Suppose that $x_i = x_{i+1}$. then the only non-zero terms are where one of x_i and x_{i+1} are omitted, hence

$$\psi'(x_1, ..., x_p) = (-1)^i \psi(x_1, ..., \hat{x_i}, ..., x_p) x_i + (-1)^{i+1} \psi(x_1, ..., \widehat{x_{i+1}}, ..., x_p) x_{i+1} = 0$$

as required. ∎

Corollary 3.5.8 (Exterior Product of Alternating Forms)
Let M be an A-module, $\theta \in L_a^1(M)$, $\psi \in L_a^{n-1}(M)$ then the function

$$\theta \wedge \psi : M^p \to A$$
$$(x_1, ..., x_n) \to \sum_{j=1}^{n} (-1)^{j+1} \psi(x_1, ..., \hat{x_j}, ..., x_p) \theta(x_j)$$

*is an alternating form. We denote this by $\theta \wedge \psi$ and refer to it as either the **wedge product** or **exterior product**, and determines a bilinear map*

$$\wedge : L_a^1(M) \times L_a^{n-1}(M) \to L_a^n(M)$$

Lemma 3.5.9
Let $f : M^n \to N$ be an alternating map then

$$f\left(x_{\sigma(1)}, ..., x_{\sigma(n)}\right) = \varepsilon(\sigma) f(x_1, ..., x_n)$$

for any permutation $\sigma \in S_n$.

Furthermore if any of the x_i are equal then $f(x_1, ..., x_n) = 0$

Proof. A permutation σ may be represented as a product of adjacent transpositions (...) therefore it's enough to demonstrate the case $\sigma = (i \quad i+1)$. This follows directly from the definition because

$$0 = f(x+y, x+y) = f(x,x) + f(y,x) + f(x,y) + f(y,y) = f(x,y) + f(y,x)$$

Suppose $x_i = x_j$, then we may apply the first result to the transposition $\sigma = (i\ j)$ to see that $f(x_1, ..., x_n) = 0$. ∎

Lemma 3.5.10
Let $\psi : M^p \to N$ be a multilinear map. Suppose $v_1, ..., v_n \in M$ and $w_1, ..., w_p \in M$ such that

$$w_i = \sum_{j=1}^{n} a_{ij} v_j \quad \forall i = 1...p \text{ and some } a_{ij} \in A$$

Then

$$\psi(w_1, ..., w_p) = \sum_{\sigma} a_{1\sigma(1)} ... a_{p\sigma(p)} \psi\left(v_{\sigma(1)}, ..., v_{\sigma(p)}\right)$$

where σ runs over all functions $[1, p] \to [1, n]$.

Lemma 3.5.11
Let $\psi : M^n \to N$ be an alternating map. Suppose $v_1, ..., v_n \in M$ and $w_1, ..., w_n \in M$ such that

$$w_i = \sum_{j=1}^{n} a_{ij} v_j \quad \forall i = 1...n \text{ and some } a_{ij} \in A$$

then

$$\psi(w_1, ..., w_n) = \sum_{\sigma \in S_n} \varepsilon(\sigma) a_{1\sigma(1)} ... a_{n\sigma(n)} \psi(v_1, ..., v_n)$$

Proof. By (3.5.10)

$$\psi(w_1, ..., w_n) = \sum_{\sigma} a_{1\sigma(1)} ... a_{n\sigma(n)} \psi\left(v_{\sigma(1)}, ..., v_{\sigma(n)}\right)$$

where σ ranges over all maps from $\{1, ..., n\}$ to itself. If σ is not a permutation, then it must not be injective and by (3.5.9) the corresponding term is zero. Therefore we may restrict to the case $\sigma \in S_n$. By the first part of (3.5.9) the result follows. ∎

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Proposition 3.5.12
Let M be a finite free A-module of rank n. Then the space of alternating forms $L_a^p(M;N) = 0$ for any integer $p > n$.

Proof. We may use (3.5.10) and observe that at least one basis element must occur at least twice in the expansion, and so is zero by (3.5.9). ∎

Proposition 3.5.13 (Existence and Uniqueness of Determinant Corresponding to a Basis)
Let M be a finite free A-module with basis $\{v_1, ..., v_n\}$. Then there is a unique alternating form $\Delta_v \in L_a^n(M)$ such that $\Delta_v(v_1, ..., v_n) = 1$.

Furthermore $L_a^n(M)$ is a free module of rank 1 with basis Δ_v.

*More precisely every $\Delta \in L_a^n(M)$ satisfies $\Delta = \Delta(v_1, ..., v_n)\Delta_v$. Further every Δ also satisifes **Leibniz' formula***

$$\Delta(x_1, ..., x_n) = \sum_{\sigma \in S_n} \varepsilon(\sigma) v_{\sigma(1)}^\vee(x_1)...v_{\sigma(n)}^\vee(x_n)\Delta(v_1, ..., v_n)$$

Proof. Let $\{v_1^\vee, ..., v_n^\vee\}$ be the canonical dual basis of $M^\vee$. Then we may use the wedge product (3.5.8) to construct Δ_v inductively

$$\Delta_v := v_1^\vee \wedge ... \wedge v_n^\vee$$

(using "right associativity" to ensure this expression is meaningful as the left operand must always be a linear functional). It is then clear that $\Delta_v(v_1, ..., v_n) = 1$. We may apply (3.5.11) to obtain Leibniz' formula for both Δ_v and Δ, which shows that $\Delta = \Delta(v_1, ..., v_n)\Delta_v$. ∎

Corollary 3.5.14 (Existence and Uniqueness of Determinant)
*For every $n \geq 1$ there is a unique $D_n \in L_a^n(A^n)$ such that $D_n(e_1, ..., e_n) = 1$, given by the **Leibniz Formula***

$$D_n(v_1, ..., v_n) = \sum_{\sigma \in S_n} \varepsilon(\sigma) v_{1\sigma(1)}...v_{n\sigma(n)}.$$

Further $L_a^n(A^n)$ is a free A-module of rank 1 generated by D_n. Explicitly every $\Delta \in L_a^n(A^n)$ satisfies

$$\Delta = \Delta(e_1, ..., e_n)D_n$$

*Further D_n satisfies the **Laplace Expansion Formula** that is for all $i, j = 1...n$ we have the relationship*

$$D_n(v_1, ..., v_n) = \sum_{j=1}^{n} (-1)^{i+j} v_{ji} D_{n-1}\Big(\pi_i(v_1), ..., \widehat{\pi_i(v_j)}, ..., \pi_i(v_n)\Big)$$

where $\pi_i : A^n \to A^{n-1}$ drops the i-th element.

Proof. The first two statements follow from (3.5.13) in the case $M = A^n$.

The function $(-1)^{i+1} D_{n-1}(\pi_i(v_1), ..., \pi_i(v_{n-1}))$ is evidentally multilinear and alternating, that is an element of $L^{n-1}(A^n)$. Forming the wedge product with $e_i^\vee$ we obtain an element Δ of $L^n(A^n)$, which has the form given by the Laplace Expansion. Furthermore

$$\Delta(e_1, ..., e_n) = (-1)^{i+1} e_i^\vee(e_i)(-1)^{i+1} D_{n-1}\Big(\pi_i(e_n), ..., \widehat{\pi_i(e_i)}, ..., \pi_i(e_n)\Big) = 1.$$

whence by (3.5.13) $\Delta = D_n$ as required. ∎

Proposition 3.5.15 (Rank is Unique)
Let M be a finite free A-module. Then the rank is unique.

Proof. This follows from (3.5.12) and (3.5.13). Explicitly if M has bases of order p and q with $p < q$, then we deduce $L_a^q(M)$ is both zero and non-zero a contradiction. ∎

Definition 3.5.16 (Determinant of a Module)
*Let M be a finite free A-module of rank n, then we say a generator for $L_a^n(M)$ is a **determinant** and Δ_v is the **determinant** corresponding to the basis $v_1, ..., v_n$.*

*The determinant for A^n corresponding to the standard basis $e_1, ..., e_n$ is called the **standard determinant** for A^n, and denoted by D_n.*

Corollary 3.5.17 (Determinant of an endomorphism)
Let M be a finite free A-module of rank n and $f \in \mathrm{End}_A(M)$ an endomorphism. Then the corresponding linear map

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$$L_a^n(f) : L_a^n(M) \to L_a^n(M)$$

satisfies

$$L_a^n(f)(\psi) = \det(f)\psi$$

for a unique $\det(f) \in A$*, which we call the* ***determinant*** *of* f*. We have the following properties*

$$\begin{aligned} \det(f \circ g) &= \det(f)\det(g) \\ \det(1_M) &= 1_A \\ \det(f) &= \Delta_v(f(v_1), ..., f(v_n)) \end{aligned}$$

for Δ_v *any generator for* $L_a^n(M)$ *corresponding to a basis* $v_1, ..., v_n$*.*

Proof. Let Δ be a generator then $L_a^n(f)(\Delta) = \Delta_v(f(v_1), ..., f(v_n))\Delta$ by (3.5.13). Clearly $\det(f) \coloneqq \Delta_v(f(v_1), ..., f(v_n))$ then satisfies the equation for all such $\psi = a\Delta$. It's unique because $L_a^n(M)$ is a free module, and the properties follow from uniqueness. ∎

We may use this to define the determinant of a matrix

Definition 3.5.18 (Determinant of a Matrix)
Let $E \in \mathrm{Mat}_{n\times n}(A)$ *then we define the* ***determinant*** *of* E *to be simply* $\det\left(\hat{E}\right)$*.*

Proposition 3.5.19 (Expressions for Matrix Determinant)
Let $E \in \mathrm{Mat}_{n\times n}(A)$ *and denote by* $v_1, ..., v_n$ *the column vectors of* E*. Then*

$$\det(E) = D_n(v_1, ..., v_n)$$

where D_n *is the standard determinant. Further there is the Leibniz Formula*

$$\det(E) = \sum_{\sigma \in S_n} \varepsilon(\sigma) \prod_{i=1}^{n} E_{i\sigma(i)}$$

and Laplace Expansion Formula

$$\det(E) = \sum_{j=1}^{n} (-1)^{i+j} E_{ij} \det\left(E_{(ij)}\right)$$

where $E_{(ij)}$ *is the* $(n-1)\times(n-1)$ *matrix with the* i*-th row and* j*-th column dropped.*

Proof. Let $e_1, ..., e_n$ be the standard basis of A^n then by definition we have $\hat{E}(e_i) = v_i$. Recall by (3.5.14) that D_n is a generator of $L_a^n(M)$ and so by (3.5.17) we have $\det\left(\hat{E}\right) = D_n(v_1, ..., v_n)$.

The Leibniz Formula and Laplace Expansion Formula also follows directly from (3.5.14). ∎

Corollary 3.5.20 (Properties of Matrix Determinant)
The determinant satisfies a number of properties

a) $\det(EF) = \det(E)\det(F) = \det(F)\det(E)$
b) $\det(I_n) = 1$
c) $\det(PEP^{-1}) = \det(E)$
d) $\det(E^t) = \det(E)$

Proof. a)-c) follow from (3.3.114) and (3.5.17). Explicitly

$$\det(EF) = D\left(\widehat{EF}\right) = D\left(\hat{E} \circ \hat{F}\right) = D\left(\hat{E}\right)D\left(\hat{F}\right) = \det(E)\det(F)$$

and

$$\det(I_n) = D\left(\hat{I}_n\right) = D(\mathbf{1}_{A^n}) = 1_A$$

d) can be shown by using Leibniz formula and observing that $\sigma \to \sigma^{-1}$ simply permutes the elements of S_n. ∎

Proposition 3.5.21
Let M *be a finite free* A*-module and* $f \in \mathrm{End}_A(M)$*. Let* $(e_1, ..., e_n)$ *be a basis and* Δ_e *the corresponding determinant. Define*

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$$f^{\mathrm{ad}}(x) \coloneqq \sum_{j=1}^{n} \Delta_e\big(f(e_1), ..., f\big(e_{j-1}\big), x, f\big(e_{j+1}\big), ..., f(e_n)\big) e_j.$$

Then

$$f \circ f^{\mathrm{ad}} = f^{\mathrm{ad}} \circ f = \det(f) \mathbf{1}_M$$

Furthermore the matrix of f^{ad} with respect to the given basis satisfies

$$\big[f^{\mathrm{ad}}\big]_{ij} = (-1)^{i+j} \det\Big(\big[f\big]_{(ji)}\Big)$$

where $M_{(ji)}$ denotes the matrix M with the j-th row and i-th column removed.

Proof. We may consider by (3.5.7) the alternating map $\psi \in L_a^{n+1}(M; M)$ given by

$$\psi(x_1, ..., x_{n+1}) \coloneqq \sum_{j=1}^{n+1} (-1)^{j+1} \Delta_e\big(x_1, ..., \hat{x}_j, ..., x_{n+1}\big) x_j$$

By (3.5.12) necessarily ψ is identically zero. Evaluating at $(x, f(e_1), ..., f(e_n))$ we find that

$$0 = \det(f) x + \sum_{j=1}^{n} (-1)^j \Delta_e\Big(x, f(e_1), ..., \widehat{f\big(e_j\big)}, ..., f(e_n)\Big) f\big(e_j\big)$$

whence

$$\begin{aligned} \det(f) x &= \sum_{j=1}^{n} (-1)^{j+1} \Delta_e\Big(x, f(e_1), ..., \widehat{f\big(e_j\big)}, ..., f(e_n) f\big(e_j\big)\Big) \\ &= \sum_{j=1}^{n} \Delta_e(f(e_1), ..., x, ..., f(e_n)) f\big(e_j\big) \end{aligned}$$

where we have used (3.5.9).

Therefore using the definition of $f^{\mathrm{ad}}(x)$ we find that $f\big(f^{\mathrm{ad}}(x)\big) = \det(f) x$ as required. For the second relation, first observe that (3.3.108)

$$x = \sum_{k=1}^{n} e_k^\vee(x) e_k$$

and so applying $f(-)$

$$f(x) = \sum_{k=1}^{n} e_k^\vee(x) f(e_k)$$

Therefore we may calculate directly

$$\begin{aligned} f^{\mathrm{ad}}(f(x)) &= \sum_{j=1}^{n} \Delta_e\big(f(e_1), ..., f\big(e_{j-1}\big), f(x), f\big(e_{j+1}\big), ..., f(e_n)\big) e_j \\ &= \sum_{j=1}^{n} \Delta_e(f(e_1), ..., f(e_n)) e_j^\vee(x) e_j \\ &= \det(f) \sum_{j=1}^{n} e_j^\vee(x) e_j \\ &= \det(f) x \end{aligned}$$

as required.

For the final statement we may assume that $M = A^n$ with $(e_1, ..., e_n)$ the standard basis and $D_n = \Delta_e$. Then by (3.5.14), (3.5.18) we have

$$\begin{aligned} \big[f^{\mathrm{ad}}\big]_{ij} &= D_n\big(f(e_1), ..., f(e_{i-1}), e_j, f(e_{i+1}), ..., f(e_n)\big) \\ &= (-1)^{i+j} D_{n-1}\Big(\pi_j(f(e_1)), ..., \pi_j\widehat{(f(e_i))}, ..., \pi_j(f(e_n))\Big) \\ &= (-1)^{i+j} \det\Big(\big[f\big]_{(ji)}\Big) \end{aligned}$$

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■

Corollary 3.5.22
Let M be a finite free A-module. Then $f \in \mathrm{End}_A(M)$ is an isomorphism if and only if $D(f) \in A^\star$.

Corollary 3.5.23
Let $E \in \mathrm{Mat}_{n\times n}(A)$ and define the adjugate matrix

$$E^{\mathrm{ad}}_{ij} := (-1)^{i+j} \det\left(E_{(ji)}\right)$$

Then

$$E^{\mathrm{ad}}E = EE^{\mathrm{ad}} = \det(E)I_n$$

Proof. By definition and (3.5.21) we have that $\widehat{E^{\mathrm{ad}}} = \hat{E}^{\mathrm{ad}}$. The identities of endomorphisms are proven in (3.5.21), and we deduce the corresponding identities of matrices by (3.3.116). ■

3.5.2 Trace Map

Proposition 3.5.24 (Existence and Uniqueness of the Trace Map)
Let M be an A-module, then there exists a well-defined A-bilinear map

$$\begin{aligned} M \otimes_A M^\vee &\to \mathrm{End}_A(M) \\ m \otimes \theta &\to \theta(-)\cdot m\,\cdot \end{aligned}$$

When M is a finite free A-module this is bijective, and there exists a unique A-linear map $\mathrm{Tr} : \mathrm{End}_A(M) \to A$ making the following diagram commute

$$\begin{array}{ccc} M \otimes_A M^\vee & \xrightarrow{\sim} & \mathrm{End}_A(M) \\ & \searrow & \downarrow \mathrm{Tr} \\ & & A \end{array}$$

where the diagonal map is simply evaluation

$$m \otimes \theta \to \theta(m)$$

Equivalently it is the unique A-linear map satisfying

$$\mathrm{Tr}(\theta(-)\cdot m) = \theta(m)$$

for all $m \in M$ and $\theta \in M^\vee$.

Finally we have the identity

$$\mathrm{Tr}(\phi) = \mathrm{Tr}([\phi]) = \sum_{i=1}^{n} [\phi]_{ii}$$

for all bases $\mathcal{B}$ of M.

Proof. There exists an obvious A-bilinear map $M \times M^\vee \to \mathrm{End}_A(M)$ so the map exists by the universal property. If M is finite free then it has a basis $\{v_1, ..., v_n\}$. By (3.3.108) $M^\vee$ has basis $\{v_1^\vee, ..., v_n^\vee\}$ and by (...) $M \otimes_A M^\vee$ has basis $\left\{v_i \otimes v_j^\vee\right\}$. By (3.3.109) this maps to a basis of $\mathrm{End}_A(M)$ and so it is an isomorphism.

Similarly the map $M \times M^\vee \to A$ is A-bilinear, and so the evaluation map $M \otimes_A M^\vee \to A$ exists. In light of the isomorphism then Tr exists and satisfies the required uniqueness property. ■

Proposition 3.5.25 (Trace is Symmetric)
Let $\phi, \psi : M \to M$ be endormophisms of a finite free A-module. Then

$$\mathrm{Tr}(\phi \circ \psi) = \mathrm{Tr}(\psi \circ \phi)$$

Proof. We may verify that the mapping

$$\begin{aligned} \mathrm{End}_A(M) \times \mathrm{End}_A(M) &\to A \\ (\phi, \psi) &\to \mathrm{Tr}(\phi \circ \psi) \end{aligned}$$

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is A-bilinear. Therefore it is sufficient to show the relationship holds for rank-one endomorphisms $\phi = \theta(-) \cdot m$ and $\psi = \iota(-) \cdot n$, as these generate $\mathrm{End}_A(M)$. Then we have

$$\mathrm{Tr}(\phi \circ \psi) = \mathrm{Tr}(\theta(\iota(-) \cdot n) \cdot m) = \mathrm{Tr}((\iota(-)\theta(n)) \cdot m) = \iota(m)\theta(n)$$

and by symmetry the result follows. ∎

Recall from (...) that every endomorphism $\phi : M \to M$ determines a family of endomorphisms $L^r_a(\phi) : L^r_a(M) \to L^r_a(M)$ for all integers $1 \leq r \leq n$, where n is the rank of M.

Proposition 3.5.26 (Determinant as a Trace)
Let M be a finite-free A module of rank n and $\phi : M \to M$ an endomorphism. Then

$$\det(\phi) = \mathrm{Tr}(L^n_a(\phi))$$

and

$$\mathrm{Tr}(\phi) = \mathrm{Tr}(L^1_a(\phi))$$

Definition 3.5.27
Let K be a subset of $[1,n]$ of order k. Then we denote by $\sigma_K : [1,k] \to K$ the unique order preserving bijection.

Denote by $\mathfrak{F}_k(n)$ the family of subsets of $[1,n]$ of order k.

If X is an $m \times n$ matrix and $K \subset [1,m]$, $H \subset [1,n]$, then we denote by $X_{H,K}$ the $\#K \times \#H$ matrix given by

$$\left(X_{H,K}\right)_{ij} = X_{\sigma_K(i),\sigma_H(j)}$$

Lemma 3.5.28
Let M be an A-module with subset $\{v_1, ..., v_n\}$ and $\psi : M^k \to N$ be an alternating map. Suppose that $x_1, ..., x_k$ satisfy

$$x_i = \sum_{j=1}^{n} A_{ji} v_j.$$

for some $n \times k$ matrix A. Then

$$\psi(x_1, ..., x_k) = \sum_{K \subset \mathfrak{F}_k(n)} \det\left(A_{K,[1,k]}\right)\psi(v_K)$$

where $v_K := \left(v_{\sigma_K(1)}, ..., v_{\sigma_K(k)}\right)$.

Proof.

$$\begin{aligned}
\psi(x_1, ..., x_k) &= \sum_{i_1,...,i_k} a_{i_1 1} ... a_{i_k k} \psi\left(v_{i_1}, ..., v_{i_k}\right) \\
&= \sum_{K \subset \mathfrak{F}_k(n)} \sum_{\sigma \in S_k} a_{\sigma_K(\sigma(1))1} ... a_{\sigma_K(\sigma(k))k} \psi\left(v_{\sigma_K(\sigma(1))}, ..., v_{\sigma_K(\sigma(k))}\right) \\
&= \sum_{K \subset \mathfrak{F}_k(n)} \sum_{\sigma \in S_k} \varepsilon(\sigma) a_{\sigma_K(\sigma(1))1} ... a_{\sigma_K(\sigma(k))k} \psi\left(v_{\sigma_K(1)}, ..., v_{\sigma_K(k)}\right) \\
&= \sum_{K \subset \mathfrak{F}_k(n)} \det\left(A_{K,[1,k]}\right)\psi(v_K)
\end{aligned}$$

∎

Proposition 3.5.29
Let M be a finite free A-module with $v_1, ..., v_n$ a basis. For every subset $K \subset [1,n]$ of order k denote

$$v^\vee_K := v_{\sigma_K(1)} \wedge ... \wedge v_{\sigma_K(k)}.$$

Then $L^k_a(M)$ has basis $\left\{v^\vee_K\right\}_{K \in \mathfrak{F}_k(n)}$. Furthermore we have the identity

$$v^\vee_K(v_{K'}) = \delta_{KK'}$$

and each basis element is uniquely determined by this relationship.

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Proposition 3.5.30
Let M be a finite free A-module with basis $\mathcal{B} = \{v_1, ..., v_n\}$. Let $\phi : M \to M$ be an A-module endomorphism. Then for all $K \in \mathfrak{F}_k(n)$

$$L_a^k(\phi)(v_K^\vee) = \sum_{H \in \mathfrak{F}_k(n)} \det\big(X_{K,H}\big) v_H^\vee$$

where $X = [\phi]_{\mathcal{B}}$.

In particular we see that

$$\mathrm{Tr}\big(L_a^k(\phi)\big) = \sum_{K \in \mathfrak{F}_k(n)} \det\big(X_{K,K}\big)$$

Proof. We have for $H \in \mathfrak{F}_k(n)$

$$\begin{aligned} L_a^k(\phi)(v_K^\vee)(v_H) &= v_K^\vee\Big(\phi\big(v_{\sigma_H(1)}\big), ..., \phi\big(v_{\sigma_H(k)}\big)\Big) \\ &= \sum_{L \subset \mathfrak{F}_k(n)} \det\big(X_{L,H}\big) v_K^\vee(v_L) \\ &= \det\big(X_{K,H}\big) \end{aligned}$$

By (3.5.29) then both sides agree on v_H for all $H \in \mathfrak{F}_k(n)$, which shows they are equal by (3.5.28). ∎

Proposition 3.5.31
Let M be a finite-free A module of rank n and $\phi : M \to M$ an endomorphism. For $a, b \in A$ we have

$$\det(a1_M + b\phi) = \sum_{k=0}^{n} \mathrm{Tr}\big(L_a^k(\phi)\big) a^{n-k} b^k$$

Proof. To calculate the left hand side consider a basis $\{v_1, ...v_n\}$ for M and the corresponding determinant Δ_v. Then

$$\det(a1_M + b\phi) = \Delta_v(av_1 + b\phi(v_1), ..., av_n + b\phi(v_n))$$

Expand to find that this equals

$$\sum_{k=0}^{n} a^{n-k} b^k \sum_{K \in \mathfrak{F}_k(n)} \Delta_v(x_{1K}, ..., x_{nK})$$

where $x_{iK} = \phi(v_i)$ if $i \in K$ and v_i otherwise.

Define τ_K to be the unique shuffle permutation of type $(k, n-k)$ (3.2.37) such that $\tau_K\,|_{[1,k]} = \sigma_K$. Denote the inner term of the sum by z_K. Then we have by (3.5.9)

$$z_K = \varepsilon(\tau_K)\Delta_v\Big(\phi\big(v_{\sigma_K(1)}\big), ..., \phi\big(v_{\sigma_K(k)}\big), v_{\sigma_H(1)}, ..., v_{\sigma_H(n-k)}\Big)$$

where H is the complement of K in $[1, n]$. Then by considering Δ_v to be an alternating form in the first k arguments, we may apply (3.5.28) to find

$$z_K = \varepsilon(\tau_K) \sum_{L \subset \mathfrak{F}_k(n)} \det\big(X_{L,K}\big)\Delta_v\Big(v_{\sigma_L(1)}, ..., v_{\sigma_L(k)}, v_{\sigma_H(1)}, ..., v_{\sigma_H(n-k)}\Big)$$

The summand is non-zero if and only if $L \cap H = \varnothing$, which occurs if and only if $L = K$. Therefore we conclude that

$$\begin{aligned} z_K &= \varepsilon(\tau_K)\det\big(X_{K,K}\big)\Delta_v\Big(v_{\tau_K(1)}, ..., v_{\tau_K(n)}\Big) \\ &= \det\big(X_{K,K}\big)\Delta_v(v_1, ..., v_n) \\ &= \det\big(X_{K,K}\big) \end{aligned}$$

The result then follows easily from (3.5.30). ∎

3.5.3 Block Matrices

Proposition 3.5.32
Let $A = \mathbb{Z}[X_{11}, ..., X_{nn}]$ be a polynomial ring in n^2 indeterminates. Consider the matrix $\big(X_{ij}\big) \in \mathrm{Mat}_{n\times n}(A)$ and let D_n denote the determinant of this matrix.

Then for any commutative ring B and matrix $M \in \mathrm{Mat}_{n\times n}(B)$ we have

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$$\det(B) = D_n(B_{11}, ..., B_{1n}, B_{21}, ..., B_{nn})$$

where the right hand side corresponds the obvious evaluation homomorphism $A \to B$.

Proposition 3.5.33 (Block Matrix Determinant)
Suppose M_{ij} $(1 \leq i \leq n, 1 \leq j \leq n)$ are $m \times m$ matrices with coefficients in a ring A and which commute pairwise. Form the matrix M with coefficients in A as follows

$$M = \begin{pmatrix} M_{11} & ... & M_{1n} \\ \vdots & & \vdots \\ M_{n1} & ... & M_{nn} \end{pmatrix}$$

Then

$$\det(M) = \det(D_n(M_{11}, ..., M_{1n}, M_{21}, ..., M_{nn}))$$

Proof. We observe without proof that there is a "block" correspondence

$$\mathrm{Mat}_{n\times n}(\mathrm{Mat}_{m\times m}(A)) \leftrightarrow \mathrm{Mat}_{(nm)\times(nm)}(A)$$

which commutes with matrix multiplication.

Let B denote the commutative subring of $\mathrm{Mat}_{m\times m}(A)$ generated by the matrices M_{ij}. Then we may regard M as an element of $\mathrm{Mat}_{n\times n}(B)$, and denote this by $\widetilde{M}$ to disambiguate. As B is commutative we my consider the adjugate matrix $\widetilde{M}^{\mathrm{ad}}$. Further consider the matrix M' given in block form by

$$M' = \begin{pmatrix} \left(\widetilde{M}^{\mathrm{ad}}\right)_{11} & 0 & ... & 0 \\ \left(\widetilde{M}^{\mathrm{ad}}\right)_{21} & I_m & ... & 0 \\ \vdots & \vdots & \ddots & \vdots \\ \left(\widetilde{M}^{\mathrm{ad}}\right)_{n1} & 0 & ... & I_m \end{pmatrix}$$

Recall by (3.5.23) and (3.5.32) that

$$\widetilde{M}^{\mathrm{ad}}_{11} = D_{n-1}(M_{22}, ..., M_{2n}, M_{31}, ..., M_{nn})$$

and

$$\widetilde{M}\widetilde{M}^{\mathrm{ad}} = \begin{pmatrix} \det\left(\widetilde{M}\right) & ... & 0 \\ \vdots & \ddots & \vdots \\ 0 & ... & \det\left(\widetilde{M}\right) \end{pmatrix}$$

as elements of $\mathrm{Mat}_{n\times n}(B)$. Therefore computing the matrix product block-wise we find

$$MM' = \begin{pmatrix} \det\left(\widetilde{M}\right) & M_{12} & ... & M_{1n} \\ 0 & M_{22} & ... & M_{2n} \\ \vdots & \vdots & & \vdots \\ 0 & M_{n2} & ... & M_{nn} \end{pmatrix}$$

Then we may show using the Leibniz Expansion that

$$\det(M) \cdot \det(M') = \det(MM') = \det\left(\det\left(\widetilde{M}\right)\right) \cdot \det\left(M_{[n+1,mn,n+1,mn]}\right)$$

By induction on n we may then assume $\det\left(M_{[n+1,mn,n+1,mn]}\right) = \det(D_{n-1}(M_{22}, ..., M_{2n}, ..., M_{nn})) = \det\left(\left(\widetilde{M}^{\mathrm{ad}}\right)_{11}\right)$. Similarly we may argue by the Leibniz expansion that $\det(M') = \det\left(\left(\widetilde{M}^{\mathrm{ad}}\right)_{11}\right)$. When this is not a zero-divisor in A then we conclude that

$$\det(M) = \det\left(\det\left(\widetilde{M}\right)\right) = \det(D_n(M_{11}, ..., M_{1n}, ..., M_{nn}))$$

as required.

In order to handle the case where $\det\left(\left(\widetilde{M}^{\mathrm{ad}}\right)_{11}\right)$ is a zero divisor we may replace the ring A with the polynomial ring $A[Z]$ and consider the matrix N for which

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$$\tilde{N} = \widetilde{M} + (Z \cdot I_m) \cdot I_n.$$

Then the submatrices of N still commute pairwise. Further $\det\Big(\big(\tilde{N}^{\mathrm{ad}}\big)_{11}\Big)$ is a polynomial of non-zero degree in Z whose leading coefficient is 1, and therefore is not a zero-divisor. We conclude from the previous case that

$$\det(N) = \det(D_n(N_{11}, ..., N_{1n}, ..., N_{nn}))$$

Finally we may apply the ring homomorphism $A[Z] \to A$ (evaluation at 0) to deduce the general case. ∎

3.5.4 Exterior Product

Proposition 3.5.34
Let M be an A-module. There is a bilinear map

$$\wedge : L_a^p(M) \times L_a^q(M) \to L_a^{p+q}(M)$$

given by

$$(f \wedge g)(x_1, ..., x_{p+q}) := \sum_{\sigma \in \mathrm{Sh}(p,q)} \varepsilon(\sigma) f\big(x_{\sigma(1)}, ..., x_{\sigma(p)}\big) g\big(x_{\sigma(p+1)}, ..., x_{\sigma(p+q)}\big)$$

where the sum is taken over shuffle permutations (3.2.37).

Proof. Suppose that $x_i = x_j$ then we require to prove that $(f \wedge g)(x_1, ..., x_{p+q}) = 0$ in order to demonstrate that it is alternating. If $1 \leq i, j \leq p$ then this follows from (3.5.9). Similarly if $p < i, j \leq p + q$. So we may consider the case $i \leq p < j$. Consider the family of right cosets $\mathrm{Sh}(p,q)/\{e, (i,j)\}$, and let Σ be some coset representatives. Then by (3.2.10)

$$\mathrm{Sh}(p,q) = \bigsqcup_{\sigma \in \Sigma} \{\sigma, (i,j)\sigma\}$$

Therefore the sum is equal to

$$\sum_{\sigma \in \Sigma} \varepsilon(\sigma) \big[f(x_\sigma) g(x_\sigma) - f\big(x_{(i\,j)\sigma}\big) g\big(x_{((i\,j)\sigma)}\big) \big]$$

so when $x_i = x_j$ this sum is zero. ∎

Proposition 3.5.35 (Basic Properties of Exterior Product)
Let M be an A-module. Suppose $f \in L_a^p(M)$, $g \in L_a^q(M)$ and $h \in L_a^r(M)$. Then the following properties hold

a) $f \wedge g = (-1)^{pq} g \wedge f$
b) $(f \wedge g) \wedge h = f \wedge (g \wedge h)$

Proof.

a) By definition and (3.2.43)

$$\begin{aligned}(f \wedge g)(x) &= \sum_{\sigma \in \mathrm{Sh}(p,q)} \varepsilon(\sigma) f\big(x_{\sigma(1)}, ..., x_{\sigma(p)}\big) g\big(x_{\sigma(p+1)}, ..., x_{\sigma(p+q)}\big) \\ &= \sum_{\sigma \in \mathrm{Sh}(q,p)} (-1)^{pq} \varepsilon(\sigma) f\big(x_{\sigma(p+1)}, ..., x_{\sigma(p+q)}\big) g\big(x_{\sigma(1)}, ..., x_{\sigma(p)}\big) \\ &= (-1)^{pq}(g \wedge f)\end{aligned}$$

b) Similarly by definition

$$\begin{aligned}((f \wedge g) \wedge h)(x_1, ..., x_{p+q+r}) &= \sum_{\sigma \in \mathrm{Sh}(p+q,r)} \varepsilon(\sigma)(f \wedge g)\big(x_{\sigma(1)}, ..., x_{\sigma(p+q)}\big) h\big(x_{\sigma(p+q+1)}, ..., x_{\sigma(p+q+r)}\big) \\ &= \sum_{\sigma \in \mathrm{Sh}(p+q,r)} \sum_{\tau \in \mathrm{Sh}(p,q)} \varepsilon(\sigma)\varepsilon(\tau) f\big(x_{\sigma(\tau(1))}, ..., x_{\sigma(\tau(p))}\big) g\big(x_{\sigma(\tau(p+1))}, ..., x_{\sigma(\tau(p+q))}\big) \\ &\qquad h\big(x_{\sigma(p+q+1)}, ..., x_{\sigma(p+q+r)}\big) \\ &\overset{(3.2.44)}{=} \sum_{\sigma \in \mathrm{Sh}(p,q,r)} \varepsilon(\sigma) f\big(x_{\sigma(1)}, ..., x_{\sigma(p)}\big) g\big(x_{\sigma(p+1)}, ..., x_{\sigma(p+q)}\big) h\big(x_{\sigma(p+q+1)}, ..., x_{\sigma(p+q+r)}\big)\end{aligned}$$

and symmetrically

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$$
\begin{aligned}
(f \wedge (g \wedge h))(x_1, ..., x_{p+q+r}) &= \sum_{\sigma \in \mathrm{Sh}(p,q+r)} \varepsilon(\sigma) f\left(x_{\sigma(1)}, ..., x_{\sigma(p)}\right)(g \wedge h)\left(x_{\sigma(p+1)}, ..., x_{\sigma(p+q+r)}\right) \\
&= \sum_{\sigma \in \mathrm{Sh}(p,q+r)} \sum_{\tau \in \mathrm{Sh}(q,r)} \varepsilon(\sigma)\varepsilon(\tau) f\left(x_{\sigma(1)}, ..., x_{\sigma(p)}\right) g\left(x_{\sigma(\tau(p+1))}, ..., x_{\sigma(\tau(p+q))}\right) \\
&\qquad h\left(x_{\sigma(\tau(p+q+1))}, ..., x_{\sigma(\tau(p+q+r))}\right) \\
&\overset{(3.2.44)}{=} \sum_{\sigma \in \mathrm{Sh}(p,q,r)} \varepsilon(\sigma) f\left(x_{\sigma(1)}, ..., x_{\sigma(p)}\right) g\left(x_{\sigma(p+1)}, ..., x_{\sigma(p+q)}\right) h\left(x_{\sigma(p+q+1)}, ..., x_{\sigma(p+q+r)}\right)
\end{aligned}
$$

therefore we conclude that the expressions are equal.

■

3.5.5 Matrix Rank

Definition 3.5.36 (Column and Row Rank)
*Let k be a field and E an $m \times n$ a matrix over k. Consider the canonical vector spaces k^n and k^m. Then define the **column rank** of E to be*

$$\mathrm{rank}_k\left(\hat{E}\right)$$

*and the **row rank** of E to be*

$$\mathrm{rank}_k\left(\widehat{E^t}\right)$$

Proposition 3.5.37 (Row Rank = Column Rank)
Let E be a matrix over k, then row rank and column rank are equal, and denote this by $\mathrm{rk}(E)$.

It is also the maximal number of linearly independent rows, or columns, and furthermore $\mathrm{rk}(E) \leq \min(m, n)$.

*We say E is **full rank** if $\mathrm{rk}(E) = \min(m, n)$.*

Proof. By (3.3.143) $\mathrm{rank}_k\left(\hat{E}\right) = \mathrm{rank}_k\left(\hat{E}^\vee\right)$ and by (3.3.120) this equals $\mathrm{rank}_k\left(\widehat{E^t}\right)$ as required.

The columns (resp. rows) clearly span $\mathrm{Im}\left(\hat{E}\right)$ (resp. $\mathrm{Im}\left(\widehat{E^t}\right)$). By (3.3.125) there are $r := \mathrm{rk}(E)$ columns (resp. rows) constituting a basis, and therefore linearly independent. For any other subset of linearly independent columns (resp. rows) we must have the order is less than r by (3.3.125). Therefore $\mathrm{rk}(E)$ is the maximal number of linearly independent rows or columns. ■

Proposition 3.5.38 (Criteria for Full Rank Square Matrix)
Let E be an $n \times n$ matrix over a field k. Then the following are equivalent

a) E is invertible
b) $\mathrm{rk}(E) = n$ (i.e. $\hat{E}$ is surjective or E is full-rank)
c) $Ev = 0 \implies v = 0$ for all column vectors v (i.e. $\hat{E}$ is injective).
d) The columns of E are linearly independent
e) $\det(E) \neq 0$

Finally E is full rank if and only if E^t is full rank.

Proof. Consider k^n with canonical basis, then by (3.3.114) E is invertible if and only if $\hat{E}$ is an isomorphism. By definition $\mathrm{rk}(E) = \mathrm{rank}_k\left(\hat{E}\right)$. Furthermore c) is equivalent to $\hat{E}$ being injective, and is also equivalent to d). Therefore the equivalence follows from (3.3.136).

Finally it's clear from either a), b) or e) that this property is self-dual. ■

Definition 3.5.39 (Minor of a matrix)
*Let E be an $m \times n$ matrix, we say a **k-minor** (for $k \leq \min(m, n)$) is the determinant of a $k \times k$ submatrix obtained by deleting $m - k$ rows and $n - k$ columns.*

Proposition 3.5.40 (Criteria for rank)
Let E be an $m \times n$ matrix over k. Then the following are equivalent

a) $\mathrm{rk}(E) \geq r$
b) There exists an $r \times r$ sub-matrix with full rank
c) There exists a non-zero r-minor

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Proof. We see b) $\iff$ c) by (3.5.38) e).

Suppose b) holds, then a-fortiori E has r linearly independent columns. Therefore by (3.5.37) $\mathrm{rk}(E) \geq r$ and a) holds.

Conversely suppose $\mathrm{rk}(E) \geq r$ then by (3.5.37) there are certainly r linearly independent columns. We consider the $m \times r$ sub-matrix E' consisting of these columns. By (3.5.37) $\mathrm{rk}(E') = r$ and there are r linearly independent rows. Choosing these rows yields an $r \times r$ submatrix E'' which has r linearly independent rows, and so by (3.5.38) is full rank as required. ∎

Corollary 3.5.41
Let E be an $m \times n$ matrix over k and r an integer. Then the following are equivalent

a) $\mathrm{rk}(E) = r$
b) r is the maximal dimension of a full-rank square sub-matrix
c) r is the maximal dimension of a non-zero minor

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